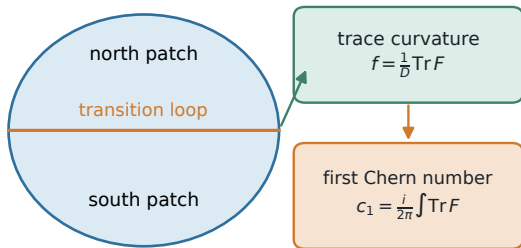


(a)

$U(1)$ trace sector

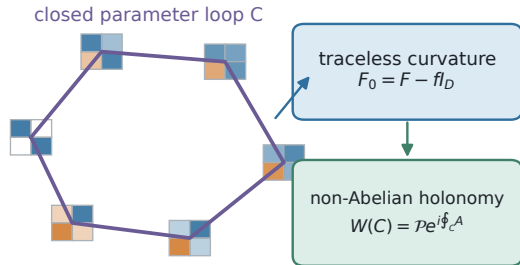


Abelian phase and determinant-line topology

[berry1984; niu1985; fukui2005]

(b)

$SU(D)$ traceless sector



Matrix transport can be noncommuting even when trace data agree

[wilczekzee1984; chen2026berry]

Topological quantization and statistical closure are separate questions.